Muon-Neutrino Disappearance in MicroBooNE and Constraints on Sterile Neutrino Oscillations

Third Year Lab Report

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The objective of this experiment was to investigate the oscillatory behaviour of neutrinos using MicroBooNE data, resulting in a reduction of the possible parameter space of the sterile neutrino. A mapping of the χ^2 of the oscillated Monte Carlo predictions with MicroBooNE data was created to visualise the parts of the parameter space that MicroBooNE data have the sensitivity to exclude. The exclusion contours created from this mapping excluded $33.1 \pm 0.9\%$ of the log-log parameter space and separated the 90% confidence region, based on the research of LSND and MiniBooNE, into two regimes: one representing a large mass squared difference with small mixing amplitudes, and the other having a smaller mass-squared difference with comparably larger mixing amplitudes.

1. INTRODUCTION

In the Standard Model, neutrinos are massless leptons that only interact with weak and gravitational forces. First proposed by Pauli in 1930 to explain the lack of energy conservation in beta decay. The Homestake experiment observed missing solar neutrinos. One proposed solution was that the neutrinos were oscillating between flavours, suggesting they have non-zero mass. The three-flavour oscillation model explains the majority of experimental results; several short-baseline neutrino experiments have reported anomalies, suggesting that the model is incomplete. [1]

One possible interpretation of these anomalies is the existence of an additional, sterile neutrino state that does not participate in weak interactions. The MicroBooNE experiment was designed to investigate the anomalies of neutrino interactions at short baselines.

2. EXPERIMENTAL APPROACH

The MicroBooNE neutrino beam is produced by Fermilab's Booster Neutrino Beamline. Protons collide with a beryllium target and generate pions, which are focused by a magnetic horn. The pions decay to produce a beam of muon neutrinos.

The simulated dataset included both external beam-off measurements, and a Monte Carlo simulation of neutrino interactions without oscillations, representing the expected signal and beam-related backgrounds. A decision tree classifier was used to separate muon-neutrino interactions from background signal, applied to both measured and simulated data.

Following background separation, a neutrino oscillation model, described in equation 1, was applied to the simulated sample to describe muon neutrino disappearance. The oscillation probability was parameterised by a mixing angle and a mass-squared difference. A χ^2 comparison between the measured data and the oscillated simulation was performed over the parameter space, allowing confidence regions for the oscillation parameters to be determined.

Using Equation 2, the muon-neutrino disappearance amplitude can be translated into an effective electron-neutrino appearance amplitude, allowing a comparison of the 90% confidence regions for sterile neutrino parameters reported by LSND and MiniBooNE with the measurements obtained from MicroBooNE data.

3. THEORY

Neutrino oscillations are a result of mixing between flavour eigenstates $(\nu_e, \nu_\mu, \nu_\tau)$ and mass eigenstates. Each flavour state is a linear combination of mass states; they are related by the Pontecorvo–Maki–Nakagawa–Sakata (PMNS) matrix [2]. As neutrinos travel, differences in their mass eigenstates lead to phase differences, causing the neutrino to change flavour over distance and energy.

The three-flavour model can be approximated by an effective two-flavour model,[3] with a single mixing angle θ. The oscillation probability between flavours ν_α and ν_β is then

$$P(\nu_\alpha \to \nu_\beta) = \sin^2(2\theta)\, \sin^2\left(\frac{1.27\,\Delta m^2 L}{E}\right), \qquad (1)$$

where Δm^2 is in eV^2, L in km, and E in GeV.

In the 3+1 model, a fourth sterile neutrino is added, which mixes with the active flavours via a 4×4 PMNS matrix, introducing a large mass-squared. The three original mass eigenstates are treated as degenerate. The effective electron-neutrino appearance angle $\theta_{\mu e}$ is related to the muon-neutrino disappearance angle $\theta_{\mu\mu}$ by

$$\sin^2(\theta_{\mu e}) = \left(1 - \sqrt{1 - \sin^2(\theta_{\mu\mu})}\right) \times 0.128, \qquad (2)$$

allowing the appearance probability to be inferred from disappearance measurements without adding extra oscillation terms.

4. RESULTS

4.1. Selection

A decision tree classifier was initially trained using 1,000 trees with a maximum depth of eight. The dataset was randomly separated into a training and testing set, with 20% of the data reserved for testing. This resulted in a training accuracy of $57.3 \pm 3.1\%$ and a testing accuracy of $54.4 \pm 0.5\%$.

The training and testing accuracies had close agreement, within random fluctuations caused by changing the seed used to split the data. This indicates that the model did not significantly overfit the data. However, the overall classification performance was poor, with the low testing accuracy implying that a substantial fraction of background events would be retained in the final event selection.

Cut Variable	Range Applied	Purity	Efficiency	Total Efficiency
trk_energy_tot	0–2 GeV	0.73	0.96	0.96
topological_score	0.4–1.0	0.88	0.61	0.59
trk_len_v	100–500 cm	0.92	0.39	0.23

TABLE I. Summary of selection cuts and resulting efficiencies.

Increasing the number of trees from 1,000 to 5,000 did not produce any statistically significant improvement in either the training or testing accuracy beyond the random fluctuations measured in the initial training.

Increasing the maximum tree depth from 8 to 16 resulted in a increase in the training accuracy to $86.4 \pm 2.2\%$, while the testing accuracy increased to $57.8 \pm 1.2\%$. The discrepancy between the training and testing accuracies suggests that the model was overfitting, learning statistical fluctuations in the training sample rather than physically significant features.

This performance can be attributed to the lack of spatial information in the input variables. The most significant distinction between cosmic muon backgrounds and muon neutrino interactions is their spatial orientation, with cosmic muons being predominantly perpendicular to the beam. As this information was not readily available to the classifier, the decision tree was unable to distinguish between these event classes reliably. Therefore, the classifier was not used as a final event selector, but instead to identify the most significant variables for selection.

4.2. Cuts

To minimise overfitting and reduce unnecessary loss of signal, the number of variables used in the cut-based selection was minimised. The first cut was applied to the reconstructed neutrino energy, as this variable is used in constructing energy histograms for comparison between simulated and measured data. Limiting the energy range removed regions with low event statistics, reducing statistical fluctuations that can arise from underpopulated bins. The decision tree classifier was used to determine the relative importance of the variables in distinguishing between muon-neutrino interactions and background events. From the feature importance analysis, the topological score and reconstructed track length were identified as the most significant variables, and cuts were therefore applied to these variables. The ranges applied, together with the remaining number of events, sample purity, selection efficiency and total efficiency after each cut, are recorded in Table I.

The topological score was used to select events whose reconstructed features are consistent with the expected topology of muon-neutrino interactions. This selection reduces contributions from events that were not correctly reconstructed and non-neutrino-like backgrounds. The reconstructed track length uses the spatial differences between beam-induced muon tracks and cosmic muon backgrounds, which are predominantly perpendicular to the beam. After the cuts, the remaining event sample contains a higher fraction of muon-neutrino interactions relative to background events. This reduction in background contribution reduces random fluctuations in the reconstructed energy spectrum that are unrelated to neutrino oscillations. As a result, the comparison between the measured data and the simulation is less influenced by non-neutrino interactions.

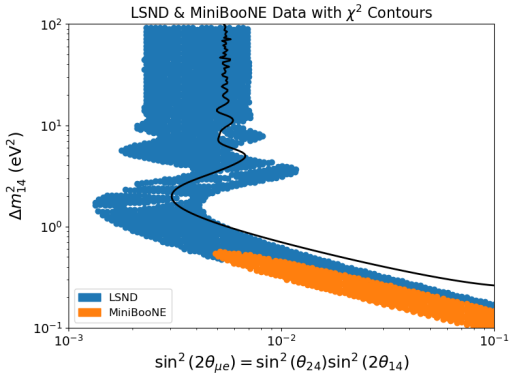

FIG. 1. Comparison of the LSND and MiniBooNE 90% confidence region and the MicroBooNE 90% confidence exclusion contour. MicroBooNE had the sensitivity to exclude the parameter space to the right, which represented $22.3 \pm 0.7\%$ of the explored parameter space.

4.3. oscillation parameters

Following the cut-based selection, the simulated dataset was weighted by the muon-neutrino survival probability, as described in Section 1. A χ^2 mapping was created over the parameter space defined by the mixing angle and mass-squared difference by comparing the oscillated simulation to the measured data. Using Equation 2, the effective muon-neutrino disappearance angle was translated into an effective electron-neutrino appearance angle, allowing the results to be interpreted within the 3+1 sterile neutrino framework. Figure reffig:preresult shows the 90% confidence region from the MicroBooNE experiment, compared to the 90% confidence regions from the LSND and MiniBooNE experiments. As shown in Figure 1, the MicroBooNE results exclude $22.3 \pm 0.7\%$ of the log–log parameter space. MicroBooNE had more sensitivity to exclude regions corresponding to smaller effective mixing angles and larger mass-squared differences.

The most significant source of uncertainty in this experiment was the approximation of a flat 15% systematic uncertainty per bin in the construction of the histograms used for the χ^2 comparison. This is the most significant source of error, as energy-dependent uncertainties associated with the reconstruction of neutrino interactions were not taken into account.

4.4. extentions

As an extension to the experiment, a more detailed evaluation of uncertainties was considered to assess the effect of replacing the flat 15% per-bin systematic uncertainty approximation. The statistical uncertainty on the measured data was treated as Poisson, with each energy bin assigned an uncertainty equal to the square root of the number of observed events, resulting in larger relative uncertainties in low-statistics bins.

To account for uncertainties in the neutrino flux prediction[4], a 10.5% uncertainty was included. To account for uncertainties in the calculation of interaction cross sections[4], a 2.73% uncertainty was added. Additionally, to account for uncertainties in detector efficiency and event selection[5], a further 6% uncertainty was applied.

Finally, a 3% energy scale uncertainty was evaluated by scaling reconstructed energies by $\pm 3\%$ and treating the resulting bin-by-bin differences as an additional uncertainty contribution[6]. All statistical and

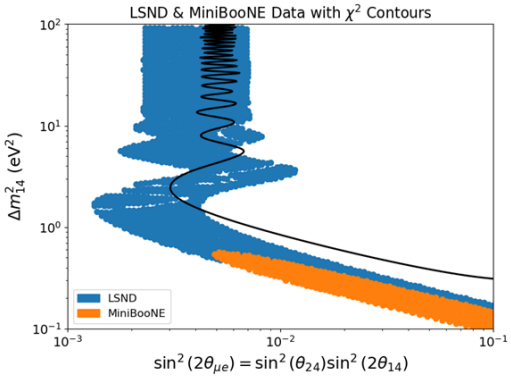

FIG. 2. Comparison of the LSND and MiniBooNE 90% confidence region and the MicroBooNE 90% confidence exclusion contour. After applying the extensions, MicroBooNE had the sensitivity to exclude the parameter space to the right, which represented $22.3 \pm 0.7\%$ of the explored parameter space.

systematic uncertainties were added in quadrature to form the total per-bin variance used in the χ^2 calculation.

The effect of including energy-dependent uncertainties is evident in Figure 2, where increased resolution is observed in the region of large mass-squared difference and small mixing angle. In contrast, this region exhibits sharp fluctuations in Figure 1, while Figure 2 displays a smoother oscillatory behaviour.

Applying cut-based selections to variables determined by the decision tree feature importance resulted in a sample purity of 92%, with a selection efficiency of 23%. While this reduced background contamination, the reduced number of events resulted in underpopulated energy bins, which reduced the sensitivity.

To minimise the loss of sensitivity, a neural network was created as an alternative. The neural network was trained using all available input variables and produced an output of the certainty that an event originated from a muon-neutrino interaction. Events with a certainty greater than 75% were selected. This increased the purity to $93.8 \pm 0.1\%$ and efficiency to $32.5 \pm 0.6\%$.

The increased signal reduced statistical fluctuations in the reconstructed energy spectrum, improving sensitivity in the parameter scan. As shown in Figure 2, $33.1 \pm 0.9\%$ of the log-log parameter space was excluded. The exclusion region separates the parameter space into two distinct regimes: a large mass-squared difference and small mixing angle, and a smaller mass-squared difference with a larger mixing angle.

5. DISCUSSION

The separation of the parameter space into two distinct regimes was not anticipated. It was expected that improvements in sensitivity or resolution of the parameter scan would result in the exclusion contour either becoming more central within the combined LSND and MicroBooNE confidence regions or excluding one of the two regimes. Instead, the increased sensitivity and resolution caused the confidence regions to shift toward the extremes of the parameter space rather than converging on a single value. This suggests that the 3+1 sterile neutrino model may be incomplete. One possible explanation is a 3+2 model, [7] in which two sterile neutrinos exist: the first with a large mass-squared difference and a low mixing angle, and the second with a small mass-squared difference and a large mixing angle.

The most significant limitation to this experimentation was removing background noise from the signal; this limitation results from the significant overlap between the reconstruction of cosmic muons and muon-neutron interactions. This could be improved by additional shielding against cosmic muons, which would enable greater certainty when classifying events and reduce signal loss.

6. CONCLUSION

This experiment has demonstrated that MicroBooNE data can provide meaningful constraints on sterile neutrino models. By applying event classification with a neural network, the experiment excluded $33.1 \pm 0.9\%$ of the log–log parameter space, separating the parameter space into two regimes: one representing a large mass squared difference with small mixing amplitudes, and the other having a smaller mass squared difference with comparably larger mixing amplitudes, providing potential evidence for a second sterile neutrino. This experiment could be extended to study additional sterile neutrino models, such as the possible existence of a second sterile neutrino within a 3+2 model. The uncertainty in the neutrino flux prediction was the dominant source of uncertainty in this experiment. As research into neutrino interactions continues, this uncertainty is expected to decrease, thereby improving the sensitivity of future analyses.

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